Machine Learning — Lecture 2

Data Understanding, Preprocessing, and Feature Engineering

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Lecture 2 at a Glance

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Data Understanding, Preprocessing, and Feature Engineering

Course Learning Outcome

CLO2: Analyze datasets and apply appropriate preprocessing, transformation, and feature engineering techniques.

Lecture purpose

- Learn how to inspect data before modeling.
- Distinguish data types, feature types, and variable roles.
- Use summary statistics and visual tools to diagnose distributions.
- Select appropriate preprocessing and feature-engineering steps.
- Prevent leakage by fitting preprocessing only on training data.

Conceptual structure

Learning phase	Core question
1. Understand the data	What does the dataset represent and what do the variables mean?
2. Explore the data	What patterns, distributions, and relationships do we observe?
3. Diagnose data quality	What problems may make learning unreliable?
4. Build the representation	How should numerical and categorical features be transformed or engineered?
5. Build a valid workflow	How do we avoid leakage and adapt preprocessing to the model?

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💡 Main idea: machine learning performance often depends as much on the **quality and representation of the data** as on the choice of model.

Learning Objectives

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By the end of Lecture 2, you will be able to:

Understanding and Exploring Data

1. **Identify** numerical, categorical, ordinal, binary, and text/image features.
2. **Describe** datasets using observations, features, targets, and metadata.
3. **Compute and interpret** mean, median, variance, standard deviation, quartiles, IQR, and outliers.
4. **Explore data** using histograms, boxplots, scatter plots, and cross-tabulations.

Preparing Data for Modeling

5. **Identify and address** missing values, duplicates, invalid entries, and inconsistent categories.
6. **Apply** scaling, normalization, standardization, encoding, and feature transformations.
7. **Explain** feature construction, feature selection, and data leakage.
8. **Design** preprocessing workflows that correctly separate training, validation, and test data.

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Lecture Roadmap

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From raw data to model-ready representation

Learning phase	Parts	Core question
1. Understand the data	**I**	What does the dataset represent, and what do the variables mean?
2. Explore the data	**II**	What patterns, distributions, spread, skew, and relationships do we observe?
3. Diagnose data quality	**III**	What problems could make learning unreliable, and how should we respond?
4. Build the representation	**IV–VI**	How do we transform numerical features, encode categories, and engineer useful features?
5. Build a valid workflow	**VII–VIII**	How do we avoid leakage and choose preprocessing that fits the learning algorithm?

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💡 **Part IX — Practice & consolidation:** diagnose feature types, detect leakage, recognize common mistakes, and justify preprocessing decisions.

Part I

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Data Understanding

Before choosing a model, we need to **understand** what the dataset **represents**, what each **variable means**, and what **problems** the data may contain.

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Why Data Understanding Comes Before Modeling

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A model learns from the representation it receives

A learning algorithm does not see the real-world problem directly.

It sees a **table**, **matrix**, **image**, **text vector**, or another representation.

Real Problem
Phenomenon in the real world

→

Data Collection
Observations are gathered

→

Representation
Features, matrix, image, or text form

→

Learning Algorithm
Model learns from the encoded input

If the **representation** is poor, the model may learn the **wrong pattern**.

Examples

Data issue	Possible consequence
Important feature missing	Model cannot use a useful signal
Wrong unit or inconsistent scale	Distance/gradient-based methods may behave poorly
Categorical value misspelled	Same category split into different artificial groups
Target information leaked into features	Unrealistically high validation/test performance
Outliers untreated without justification	Model may be overly influenced by rare observations

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💡 Good modeling begins with asking: **What exactly does each column mean, and can the model safely use it?**

Dataset Anatomy

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Observations, variables, features, targets

A standard supervised-learning dataset is often represented as:

$$X \in \mathbb{R}^{n \times p}, \qquad y \in \mathbb{R}^{n}$$

where:

- n is the number of observations;
- p is the number of features;
- each row is one example;
- each column is one measured attribute;
- y is the target variable.

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Example — Student-performance prediction

Student	Study Hours	Attendance	Previous GPA	Learning Mode	Final Result
S1	12	92%	3.4	Group	Pass
S2	4	65%	2.1	Individual	Fail
S3	9	80%	2.8	Online	Pass

Here, `Study Hours`, `Attendance`, `Previous GPA`, and `Learning Mode` are features. `Final Result` is the target.

Feature Types

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Not all variables behave the same way

Feature type	Example	Typical treatment
Continuous numerical	income, temperature, exam score	scaling, transformations, outlier analysis
Discrete numerical	number of orders, absences	may be scaled or treated as count data
Binary	has loan: yes/no	encode as 0/1
Nominal categorical	city, product type	one-hot or other categorical encoding
Ordinal categorical	low/medium/high	ordered encoding when order is meaningful
Date/time	timestamp, semester	extract useful components
Text	review, complaint	tokenize and vectorize
Image	handwritten digit	represent as pixel values or visual features

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💡 The model does not automatically know the meaning of a variable. The representation must preserve the meaning needed for learning.

Variables Have Roles

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A column’s role depends on the learning task

Consider a dataset of **bank customers**:

Customer	Age	Income	Existing Loan	Defaulted
C1	24	45000	Yes	No
C2	52	98000	No	No
C3	36	30000	Yes	Yes

Default Prediction

Supervised learning

- **Features:** Age, income, existing loan.
- **Target:** Defaulted.
- **Goal:** Predict whether a customer will default.

Customer Segmentation

Unsupervised learning

- **Features:** Age, income, existing loan.
- **Target:** None.
- **Goal:** Discover groups of similar customers.

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💡 **Key idea.** The same dataset can support **different learning tasks**. A column’s role depends on the **prediction or analysis objective**.

Data Dictionary

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Understanding what each variable represents

A **data dictionary** documents the meaning, format, and expected values of each variable.

Field	What should be documented
Name	Column name in the dataset
Meaning	Real-world quantity or concept represented
Type	Numerical, categorical, date, text, image, etc.
Unit	SAR, minutes, cm, percentage, count
Valid values	Expected range or allowed categories
Missing values	Meaning of blanks, zeros, or special codes
Timing	When the value becomes available

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Why does timing matter?

A feature that becomes available **only after the prediction must be made** may cause **data leakage** if used during model development.

> **Important.** Every feature must be **available at prediction time**. Otherwise, evaluation results may look excellent while real-world predictions fail.

First Questions to Ask About a Dataset

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A practical data-inspection checklist

Structure and Meaning

1. What does **each row** represent?
2. What does **each column** mean?
3. What is the **prediction or discovery task**?
4. Which variable, if any, is the **target**?
5. What are the **feature types** (numerical, categorical, ordinal, text, image)?

Quality and Reliability

6. Are there **missing, invalid, duplicate, or inconsistent values**?
7. Are there **outliers or extreme values**?
8. Are the **units and scales** appropriate?
9. When was each feature **collected or made available**?
10. Could any feature cause **data leakage**?

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💡 **Key idea.** Careful data inspection helps identify **quality problems, inappropriate features, and leakage risks** before modeling begins.

Part II

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Exploratory Data Analysis

Exploratory Data Analysis helps us **summarize, visualize,** and **question** the data **before fitting a model**.

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Descriptive Statistics

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Center and spread

Given values $x_1, x_2, \ldots, x_n$

The sample mean is $\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$

The sample variance is $s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$

The sample standard deviation is $s = \sqrt{s^2}$

Interpretation

Statistic	Meaning
Mean	**Average** or balance point of the values
Median	**Middle value** after sorting
Variance	Measures spread using **squared deviations**
Standard deviation	Measures **variation around the mean** in the original units

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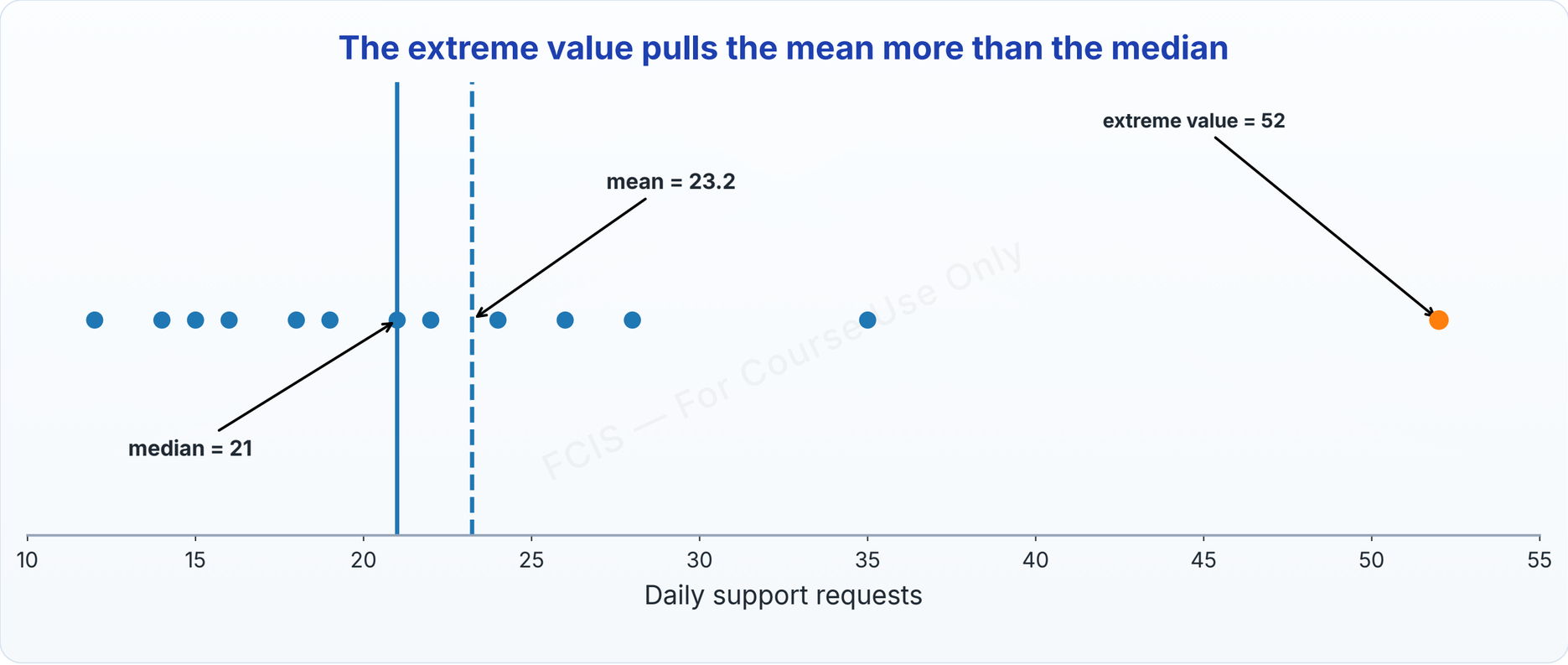
💡 **Key idea.** The **mean** is sensitive to extreme values. The **median** is often more robust when the distribution is skewed.

Mean vs. Median

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Why robust summaries matter

The **mean** uses all numerical values and is sensitive to extreme observations. The **median** is the middle value after sorting and is more robust to outliers.



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💡 **key idea.** When a distribution contains extreme values, the **median** often gives a more stable description of the center than the **mean**.

Quartiles and Interquartile Range

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A compact description of distribution spread

Quartiles divide ordered data into four parts

Outside the fences, observations are often flagged for further inspection.



interquartile range (IQR) = Q3 − Q1

The IQR summarizes the spread of the middle 50% of the data.

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> The **interquartile range** is robust to extreme values because it focuses on the middle half of the distribution.

Worked Example: Quartiles and Outliers

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Daily support requests

Sorted data: $12,\ 14,\ 15,\ 16,\ 18,\ 19,\ 21,\ 22,\ 24,\ 26,\ 28,\ 35,\ 52$

Quartiles

There are 13 observations, so the median is the 7th value:

$$Q_2 = 21$$

Lower half: $12,\ 14,\ 15,\ 16,\ 18,\ 19$

Therefore:

$$Q_1 = \frac{15 + 16}{2} = 15.5$$

Upper half: $22,\ 24,\ 26,\ 28,\ 35,\ 52$

Therefore:

$$Q_3 = \frac{26 + 28}{2} = 27$$

IQR and outlier fences

$$IQR = Q_3 - Q_1 = 27 - 15.5 = 11.5$$

Lower fence:

$$Q_1 - 1.5(IQR) = 15.5 - 17.25 = -1.75$$

Upper fence:

$$Q_3 + 1.5(IQR) = 27 + 17.25 = 44.25$$

So:

- (12>-1.75): no lower outlier.
- (35<44.25): 35 is still within the upper fence.
- (52>44.25): 52 is a potential outlier.

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Result: (52) is flagged as a potential outlier. The value (35) is the largest observation still inside the fences.

Important: the fences (-1.75) and (44.25) are **cutoff thresholds**, not necessarily whisker locations. In the corresponding boxplot, the whiskers end at the most extreme observed values within the fences: (12) and (35).

Boxplots

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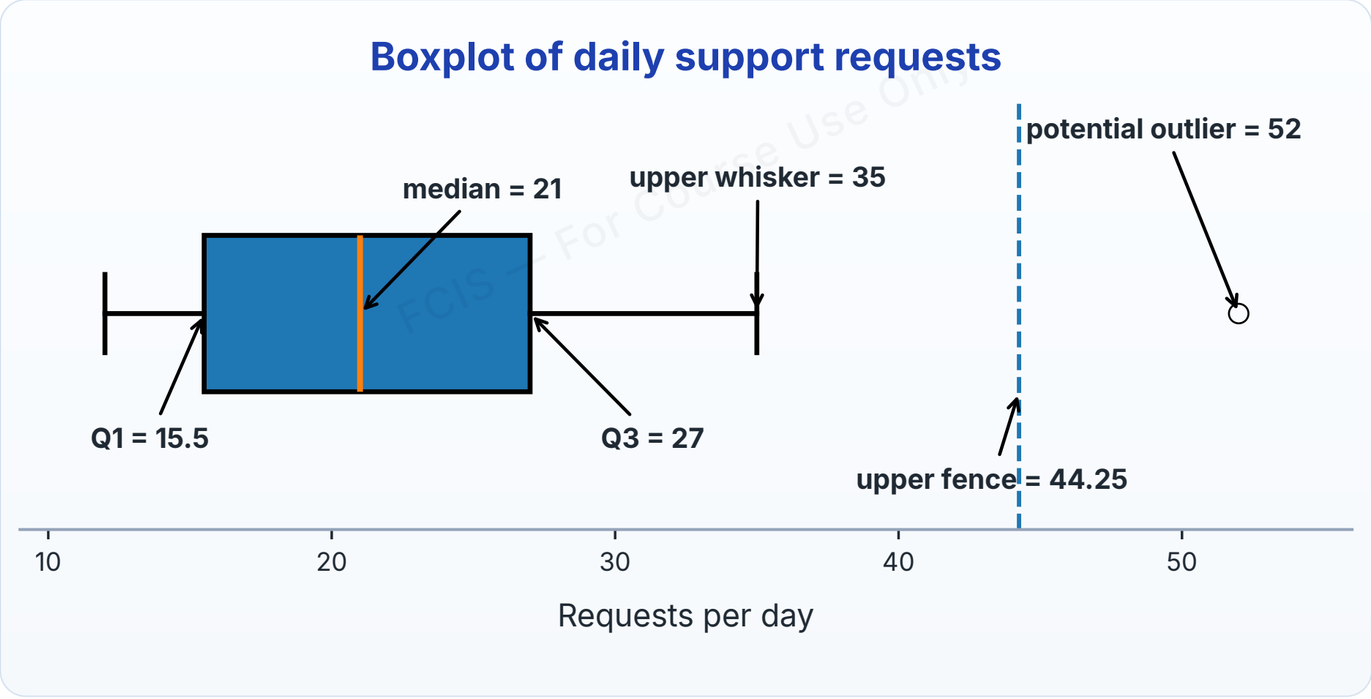
A visual summary of distribution

A **boxplot** summarizes a distribution with:

- the **box** from Q_1 to Q_3,
- a line for the **median**,
- **whiskers** extending to non-outlying observations,
- and separate marks for **potential outliers**.

In this example, the high value **52** appears beyond the upper whisker.

> 💡 Use boxplots to quickly compare center, spread, skew, and unusual observations.



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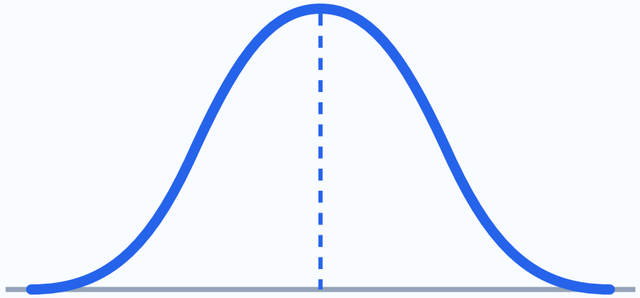
Interpreting Boxplots

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Match the boxplot to the distribution shape

Step 1 — symmetric distribution

Density curve



balanced around the center

Corresponding boxplot



median near center
whiskers similar in length

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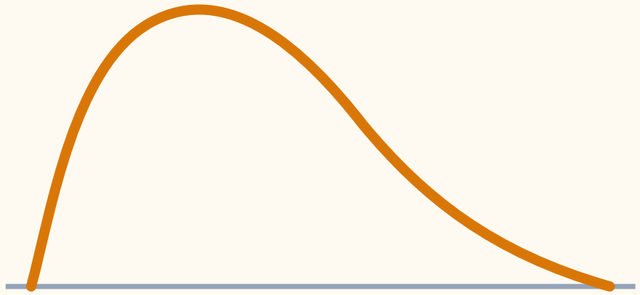
A symmetric boxplot usually has a centered median and similar spread on both sides.

A boxplot does not display the full density, but it can reveal **center**, **spread**, **asymmetry**, and potential **outliers**.

Match the boxplot to the distribution shape

Step 2 — right-skewed distribution

Density curve



Corresponding boxplot



median closer to Q1
upper side is more stretched

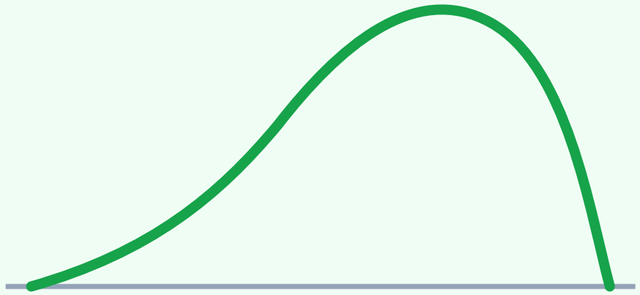
Right skew appears as more spread toward larger values; the upper whisker is often longer.

A boxplot does not display the full density, but it can reveal **center**, **spread**, **asymmetry**, and potential **outliers**.

Match the boxplot to the distribution shape

Step 3 — left-skewed distribution

Density curve



long left tail

Corresponding boxplot



median closer to Q3
lower side is more stretched

Left skew appears as more spread toward smaller values; the lower whisker is often longer.

A boxplot does not display the full density, but it can reveal **center**, **spread**, **asymmetry**, and potential **outliers**.

Match the boxplot to the distribution shape

Step 4 — compare the three patterns

Symmetric



balanced spread

Right-skewed



more spread to the right

Left-skewed



more spread to the left

Interpret the median position and relative spread together; no single visual cue is sufficient by itself.

A boxplot does not display the full density, but it can reveal **center**, **spread**, **asymmetry**, and potential **outliers**.

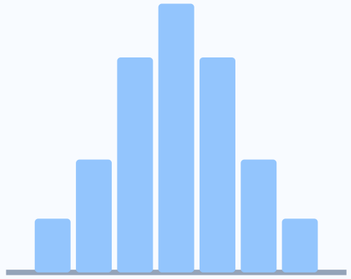
Histograms and Distribution Shape

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Seeing Frequency Patterns

A **histogram** shows how numerical values are distributed across intervals (bins).

Symmetric



Balanced around the center

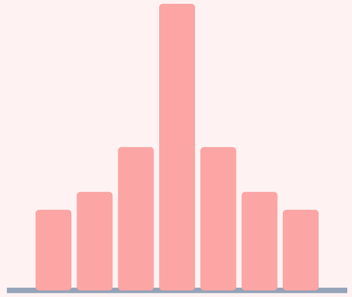
Right-skewed

Long tail toward larger values

Multimodal

Multiple peaks or groups

Heavy-tailed



More frequent extreme values

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💡 **Key idea.** Histograms reveal **distribution shape, skewness, spread**, and possible **subgroups**.

Example. Income, spending, and waiting times are often **right-skewed**. A **log transformation** may help make such features easier to model.

Scatter Plots

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Inspecting Relationships Between Two Variables

A **scatter plot** shows how two numerical variables vary together and helps identify important patterns.

Relationships

- **Linear** trends
- **Nonlinear** trends
- **Clusters** or subgroups

Unusual Patterns

- **Outliers**
- **Changing variability** (heteroscedasticity)
- **Unusual groups** of observations

Example Questions

Question	What to look for
Does study time relate to exam score?	**Positive association**
Does rent increase linearly with area?	**Linear or nonlinear trend**
Are there distinct populations?	**Clusters or subgroups**
Does the spread increase with the feature?	**Changing variability**

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Important. A visual association between two variables does **not establish causation**. Other factors may explain the observed relationship.

Correlation

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Measuring Linear Association

The **Pearson correlation coefficient** (r) measures the **direction and strength of linear association** between two numerical variables.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

How to interpret r

Value of r	Interpretation
Close to $+1$	**Strong positive** linear association
Close to 0	**Weak or no linear** association
Close to -1	**Strong negative** linear association

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Important: Correlation ranges from -1 to $+1$. The **sign** indicates direction, while the **magnitude** indicates linear strength.

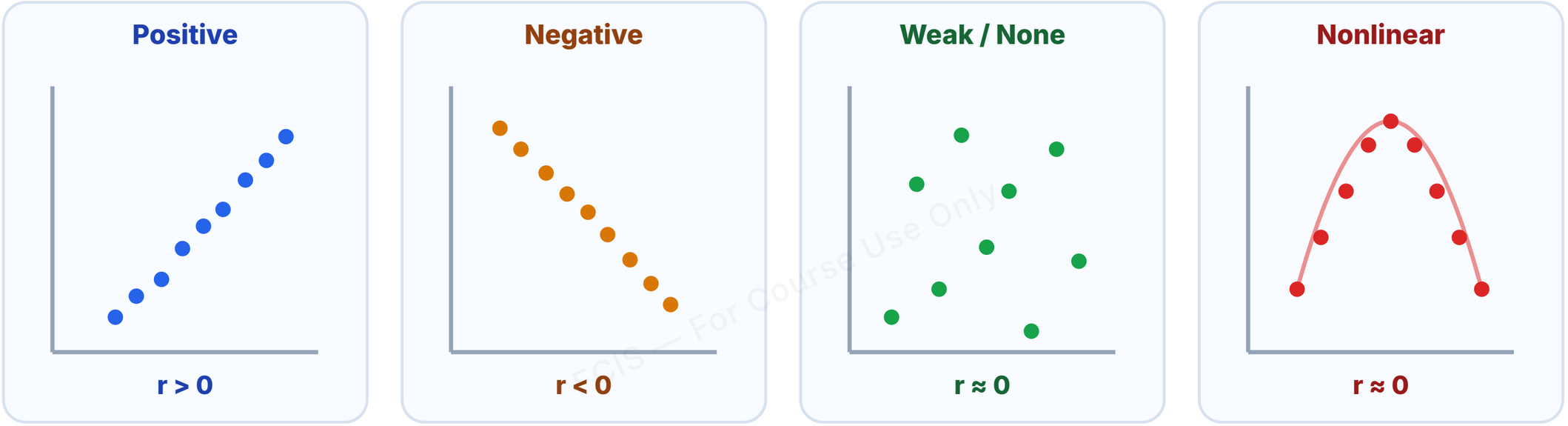
> **Correlation measures linear association only.** A value near zero can occur even when a **strong nonlinear relationship** exists. Always inspect the **scatter plot**.

Interpreting Correlation Patterns

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Different relationships can have similar correlations

Pearson correlation describes the direction and strength of a **linear relationship**, but it does not capture every pattern in the data.



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Important. A correlation near zero means **little or no linear association**, not necessarily no relationship. The last two plots have similar correlation values but **very different patterns**.

Categorical Association

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Cross-tabulations and independence

For categorical variables, a **contingency table** counts combinations of categories.

Example:

Study Method	Low Performance	High Performance	Total
Individual	18	42	60
Group	10	50	60
Online	32	28	60
Total	**60**	**120**	**180**

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If two variables are independent, the expected count is: $E_{ij} = \frac{(\text{row total})(\text{column total})}{\text{grand total}}$

The Chi-square statistic is: $\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$

💡 This test helps decide whether the observed differences are stronger than what independence would suggest.

Worked Example: Expected Counts

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Study method and performance

For the first row and first column: $E_{\text{Individual, Low}} = \frac{60 \times 60}{180} = 20$

For the first row and second column: $E_{\text{Individual, High}} = \frac{60 \times 120}{180} = 40$

Since all row totals are 60, the expected counts are:

Study Method	Expected Low	Expected High
Individual	20	40
Group	20	40
Online	20	40

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Contribution example

For Online-Low: $\frac{(32-20)^2}{20} = \frac{144}{20} = 7.2$

> Large differences between observed and expected counts contribute strongly to χ^2.

Worked Example: Chi-Square Statistic

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Summing the contributions

Each cell contributes $\frac{(O_{ij} - E_{ij})^2}{E_{ij}}$:

Cell	O	E	Contribution
Individual–Low	18	20	0.2
Individual–High	42	40	0.1
Group–Low	10	20	5.0
Group–High	50	40	2.5
Online–Low	32	20	7.2
Online–High	28	40	3.6
Total			**18.6**

Chi-Square

$$\chi^2 = 0.2 + 0.1 + 5.0 + 2.5 + 7.2 + 3.6 = 18.6$$

Degrees of freedom

$$df = (r - 1)(c - 1) = (3 - 1)(2 - 1) = 2$$

Decision

Critical value at $\alpha = 0.05$: $\chi^2_{0.05,\,2} = 5.99$

Since $18.6 > 5.99$, we **reject independence**.

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💡 Study method and performance are **not independent** ($\chi^2 = 18.6$, $df = 2$, $p < 0.001$). The association is driven mainly by the **Low-performance** column — especially **Online** (7.2) and **Group** (5.0).

Part III

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Data Quality Problems

Real datasets are imperfect. Preprocessing begins by distinguishing **data errors** from **meaningful but difficult observations**.

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Missing Values

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Why a value may be absent

Missing values can have different meanings and may require different treatments.

Situation	Example	Possible treatment
Not recorded	Age field left blank	Impute or remove
Not applicable	Spouse income for unmarried person	Explicit category or indicator
Sensor failure	Temperature reading lost	Impute using time or context
Informative missingness	High-income users skip income question	Preserve missingness information

Common strategies

- **Remove rows** when missingness is limited and removal is unlikely to introduce bias.
- **Remove features** with excessive missingness when justified.
- **Impute values** using the mean, median, mode, or a learned method.
- Add a **missingness indicator**.
- Use models that **handle missing values** directly.
- Fit imputation rules using **training data only**.

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💡 **Key idea.** The appropriate treatment depends on **why values are missing**, not just how many are missing.

Duplicates and Invalid Records

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Not every row should be trusted

Duplicate Records

- An **accidental duplicate**.
- A **legitimate repeated transaction**.
- A **repeated measurement**.
- One person recorded under **different IDs**.

Invalid or Inconsistent Values

Feature	Example
Age	**-3**
Attendance	**140%**
Exam score	**125 out of 100**
Date	Future date for a past event
Category	`M`, `Male`, `male`, `MALE` as separate entries

The last row illustrates **inconsistent category labels**, not necessarily invalid observations.

Important. Data cleaning must be **justified and documented**. Removing legitimate observations can be as harmful as retaining incorrect records.

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Outliers

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Error, rare event, or important signal?

An **outlier** is an observation unusually far from the general pattern of the data.

Possible Interpretations

Interpretation	Example
Data entry error	Age entered as 220
Measurement error	Sensor spike
Rare real event	Unusually high hospital demand
Different population	Corporate vs. individual customer
Important signal	Unusually large fraudulent transaction

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Possible Responses

- **Verify and correct** genuine errors.
- **Keep** valid and important observations.
- **Cap or winsorize** extreme values when justified.
- **Transform** highly skewed features.
- Use **robust statistics or models**.
- **Analyze separately** if a distinct group is present.

Important. Outliers are **not automatically errors**. Their treatment should reflect the data, prediction goal, and domain context.

Inconsistent Categories

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Small text differences can create artificial groups

Consider a feature named `City` containing:

```
Madinah
madinah
Al Madinah
Medina
MADINAH
```

These values may refer to **the same city**, but a model can treat them as **five separate categories**.

Standardization Steps

- **Trim** leading and trailing spaces.
- **Standardize** capitalization.
- **Map synonyms** to a consistent label.
- **Correct** spelling variants.
- **Merge rare categories** only when justified.

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> 💡 **Key idea.** Category cleaning is **not merely cosmetic**. Inconsistent labels can create artificial groups and change the **feature representation** learned by the model.

Data Leakage

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When unavailable information enters the model

Data leakage occurs when model development or evaluation uses information that should not be available for the prediction being made.

Common Examples

Leakage example	Why it is a problem
Using `final_grade` to predict course pass/fail	The feature **reveals the target**
Scaling the full dataset before splitting	**Test information** influences preprocessing
Selecting features using test performance	The test set becomes part of **model selection**
Using future transaction status to predict fraud	Information is available **only after prediction time**

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How to prevent leakage

- **Split data before fitting preprocessing steps**.
- Use **training data** to learn transformations.
- Use **validation data** for model selection.
- Keep the **test set independent** until final evaluation.

Important. Leakage can create **misleadingly high evaluation scores** that do not reflect performance on genuinely unseen data.

Part IV

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Preprocessing Numerical Features

Numerical preprocessing transforms raw features into representations that help learning algorithms **train reliably and make meaningful predictions**.

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Feature Scaling

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Why feature scale matters

Some algorithms are sensitive to **differences in feature scale**.

Examples

Feature	Typical range
Annual income	20,000–300,000
Age	18–80
Number of dependents	0–8

Without scaling, **income may dominate distance calculations** simply because its numerical range is much larger.

Which methods are affected?

Distance-based methods

- K-Nearest Neighbors (KNN)
- K-Means clustering

Optimization and regularization

- Gradient-based training
- Neural networks
- Ridge and Lasso regression

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Important. For **KNN and K-Means**, unscaled features can significantly change which observations are considered **nearest neighbors or members of the same cluster**.

Standardization

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Center and scale by standard deviation

Standardization converts a numerical feature into a z-score:

$$z = \frac{x - \mu}{\sigma}$$

where μ is the **mean** and σ is the **standard deviation**.

After standardization

- The training feature has a **mean near 0**.
- Its standard deviation is **approximately 1**.
- Features with different units become more **comparable in scale**.

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Common applications

- Ridge and Lasso regression
- Logistic regression
- K-Nearest Neighbors
- K-Means clustering
- Neural networks
- Gradient-based optimization

Prevent leakage. Estimate μ and σ using **training data only**, then apply those values to validation and test data.

Min-Max Normalization

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Rescale features to a fixed interval

Min-max scaling transforms numerical values into a range such as $[0, 1]$:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}$$

Advantages

- Produces a **consistent numerical range** for training data.
- **Preserves the ordering** of observations.
- Useful when **bounded input scales** are preferred.

Important limitations

- **Sensitive to outliers:** Extreme values determine the scaling range.
- **Unseen values:** Validation or test observations can fall **outside** $[0, 1]$.
- Requires special handling when $x_{\max} = x_{\min}$.

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💡 **Key idea. Min-max scaling** rescales training values to a fixed interval, while **standardization** centers and scales using the mean and standard deviation. Both are sensitive to **outliers**, motivating alternatives such as **robust scaling**.

Robust Scaling

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Scaling with the median and IQR

Robust scaling uses the **median** and **interquartile range (IQR)** instead of the mean and standard deviation:

$$x' = \frac{x - \text{Median}}{Q_3 - Q_1}$$

Worked Example

Suppose **Median = 30**, $Q_1 = 20$, and $Q_3 = 40$.

For $x = 50$:

$$x' = \frac{50 - 30}{40 - 20} = \boxed{1}$$

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When is it useful?

- When features contain **extreme values or outliers**.
- When the **mean and standard deviation** are strongly affected by those values.

💡 **Key idea.** Robust scaling is **less sensitive to outliers**, but it is **not always the best choice**. Select a scaling method based on the **data distribution, learning algorithm, and validation performance**.

Transforming Skewed Features

FCIS Faculty Use Only

Making numerical patterns easier to model

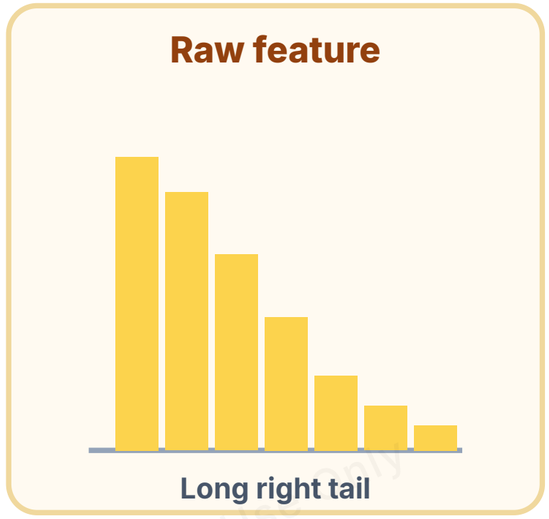
Highly **right-skewed features** may contain a few values much larger than the rest.

For nonnegative values, a common transformation is:

$$x' = \log(1 + x)$$

Possible benefits

- **Compresses large values**.
- Often **reduces right-skewness**.
- Can make relationships easier for some models to learn.





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The transformation **preserves ordering** but changes the distances between values.

💡 **Key idea.** Log transformations can reduce the influence of **very large values** and reveal useful patterns. However, they do **not guarantee a symmetric distribution**.

Binning and Discretization

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Turning numerical values into intervals

Binning converts a numerical feature into **categories defined by intervals**.

Example: Age groups

Age	Assigned category
19	Young
34	Adult
67	Senior

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Possible Benefits

- **Simplifies** complex relationships.
- Improves **interpretability**.
- Can reduce sensitivity to **small measurement differences**.
- Supports **domain-specific rules**.

Possible Limitations

- **Loses numerical information**.
- Bin boundaries may be **arbitrary**.
- Creates **artificial discontinuities**.
- Different bin choices can change results.

Important. Binning may improve interpretability, but it can discard useful predictive information. Choose interval boundaries carefully and assess their impact using **validation data**.

Part V

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Encoding Categorical Features

Most machine-learning algorithms require **numerical inputs**. Encoding converts **categorical features** into numbers while preserving their **meaning**.

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Label Encoding

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Useful only when order is meaningful

Ordinal categories — meaningful order

Consider three satisfaction levels:

$$\text{Low} \to 0, \qquad \text{Medium} \to 1, \qquad \text{High} \to 2$$

This encoding preserves the **natural order** of the categories, but does not necessarily mean the differences between levels are equal.

Nominal categories — no natural order

Consider three colors:

$$\text{Red} \to 0, \qquad \text{Blue} \to 1, \qquad \text{Green} \to 2$$

This encoding introduces an **artificial order and numerical distance** between categories.

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> 💡 **Takeaway.** Use ordinal encoding when categories have a meaningful order. For nominal categories, prefer an encoding such as **one-hot encoding** when the model would otherwise treat numerical codes as ordered or metric.

One-Hot Encoding

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Representing nominal categories without ordering

Consider **Learning Mode**, a nominal feature with three categories: **Individual**, **Group**, and **Online**.

One-hot encoding creates a separate **binary feature** for each category:

Learning Mode	Individual	Group	Online
Individual	1	0	0
Group	0	1	0
Online	0	0	1

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Advantages

- Avoids **artificial numerical ordering**.
- Produces **interpretable features**.
- Works with many standard models.

Limitations

- **Increases dimensionality**.
- Rare categories create **sparse features**.
- **Unseen categories** need a handling strategy.

💡 **Key idea.** One-hot encoding represents nominal categories **without implying that one category is greater than another**.

High-Cardinality Categories

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When one-hot encoding creates too many features

A categorical feature has **high cardinality** when it contains **many distinct values**.

Examples: Product ID, user ID, postal code, course code, and device ID.

Possible Strategies

Simpler approaches

- **Group rare categories**.
- Apply **domain-based grouping**.
- Use **frequency encoding** when appropriate.

Advanced approaches

- Use **target encoding** with leakage-safe validation.
- Learn **embeddings** in advanced models.
- **Exclude identifiers** that add no useful information.

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Important. High-cardinality features, especially **nearly unique identifiers**, can encourage models to **memorize training examples** instead of learning general patterns.

Text, Images, and Structured Inputs

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Different data require different representations

Not all input data arrive as numerical or categorical columns. **Feature extraction and engineering** can convert complex data into representations suitable for learning.

Raw data	Possible representation	Later course connection
Text reviews	Bag-of-words counts	NLP and language models
Images	Pixel intensities	Computer vision and neural networks
Time series	Lagged values, rolling statistics	Forecasting
Transaction history	Aggregated behavior features	Fraud and customer modeling
Graph data	Node and edge features	Network analysis

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Example. A transaction history can be converted into features such as **number of purchases in the past 30 days**, **average transaction amount**, and **days since the last purchase**.

Part VI

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Feature Engineering

Feature engineering creates, transforms, or selects features to help models **capture meaningful patterns** in the data.

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Feature Construction

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Creating useful variables from existing ones

Feature construction combines or transforms existing data into new, potentially more informative variables.

Examples

Raw variables	Engineered feature
Purchase dates	Days since last purchase
Total spending and number of orders	Average order value
Height and weight	Body-mass index (BMI)
Timestamp	Hour, day of week, semester
Text message	Number of exclamation marks
Transaction sequence	Failed attempts in the past hour

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💡 **Key idea.** Good features often capture **domain knowledge** that a model might struggle to discover from raw data. They must also be **available at prediction time**.

Interaction Features

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When the effect of one feature depends on another

An **interaction** occurs when the relationship between one feature and the target depends on another feature.

Examples

- **Study hours** may affect exam scores differently depending on **attendance**.
- **Loan amount** may affect default risk differently depending on **income**.
- **Age** may relate to health outcomes differently depending on **medical conditions**.

Creating an interaction feature

A simple interaction is the **product of two features**:

$$\boxed{x_{\text{interaction}} = x_1 x_2}$$

Including x_1, x_2, and $x_1 x_2$ allows a linear model to represent **effects that depend on both variables**.

> 💡 **Key idea.** Interaction features increase a model's **flexibility** without requiring a different learning algorithm.

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Polynomial Features

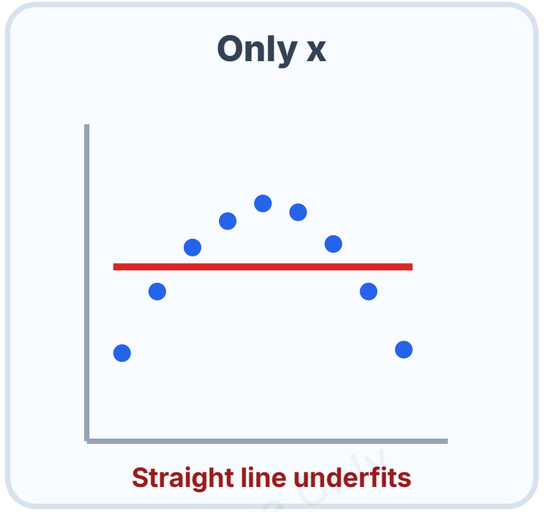
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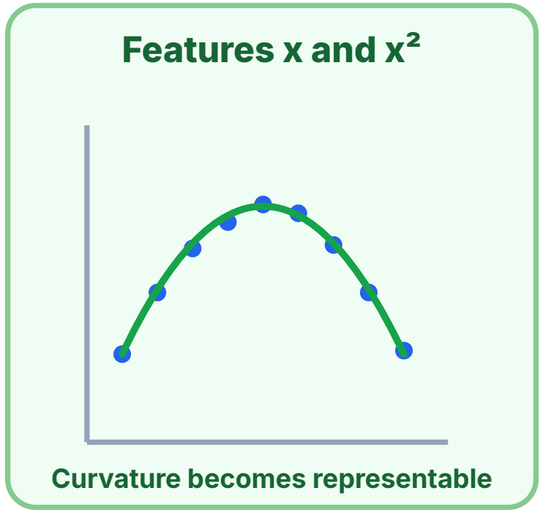
Representing nonlinear relationships

A model using only x produces a straight-line relationship.

Adding polynomial features such as x^2 or x^3 allows the model to represent **curvature**:

$$\hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2$$





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- The model remains **linear in its coefficients**, even though its predictions can follow a curve.
- This is a change in **feature representation**, not the learning algorithm.

Important. Polynomial features increase **model flexibility**, but adding too many terms may lead to **overfitting**. Choose the degree using validation data.

Feature Selection

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Keeping the features that matter

Feature selection identifies a useful subset of available features while excluding unnecessary or problematic ones.

Why Select Features?

- **Reduce overfitting**.
- Improve **interpretability**.
- Reduce **computational cost**.
- Remove **redundant variables**.
- Exclude **leakage-prone features**.
- Simplify **deployment**.

Warning Signs

- **Nearly constant** values.
- **Duplicate or redundant** information.
- Available **only after prediction time**.
- **Excessive missingness**.
- Identifiers with little generalizable meaning.
- Suspiciously strong relationships with the target.

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Important. A feature that produces unusually high performance may be genuinely informative—or it may contain **leaked target information**. Investigate before using it.

Feature Engineering vs. Model Choice

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Representation and algorithm work together

Different algorithms have different needs for **feature representation**.

Model	Role of feature engineering
Linear / logistic regression	Explicit nonlinear and interaction features may help
KNN / K-Means	Feature scaling strongly influences distance calculations
Decision trees	Generally do not need scaling, but irrelevant features can hurt
Neural networks	Can learn representations, but benefit from suitable preprocessing
Text / image models	Appropriate representations are central to learning

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💡 **Key idea.** Good performance depends on both the **feature representation** and the **learning algorithm**. A more complex model cannot reliably compensate for poor-quality features or data leakage.

Part VII

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Building a Leakage-Safe Preprocessing Workflow

To prevent **data leakage**, transformations that learn from data must be fitted using the **training set only**.

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The Correct Development Workflow

For Course Use Only

Fit transformations on training data only

Correct workflow

1. **Split** the dataset into training, validation, and test sets.
2. **Fit preprocessing** using training data only.
3. **Transform training data** using the fitted parameters.
4. **Apply the same transformations** to validation data.
5. **Select the model** using validation performance.
6. **Evaluate once** on the held-out test set.

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Example: Standardization

Estimate the mean and standard deviation from training data:

$$z = \frac{x - \mu_{\text{train}}}{\sigma_{\text{train}}}$$

Use these **same training-set values** when transforming validation and test observations.

> **Important.** Validation and test data must represent **unseen observations**. They must not influence the fitting of preprocessing transformations.

Incorrect Workflow: Leakage

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When preprocessing uses test information

Incorrect workflow

1. Compute **preprocessing statistics** using the entire dataset.
2. Split the data into training and test sets.
3. Train the model.
4. Report the **test performance**.

Why is this incorrect?

- The **test distribution** has already influenced preprocessing.
- The evaluation is no longer fully **independent**.
- Even without using target labels, information from test observations has entered model development.

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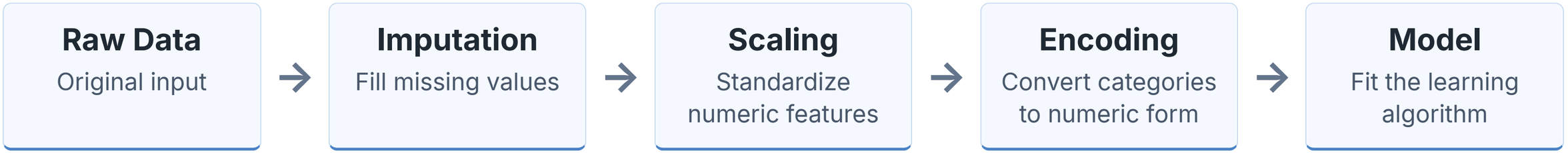
Common leakage risks: Imputation, scaling, feature selection, and target encoding must be handled carefully. Fit data-dependent transformations on **training data only**.

Preprocessing Pipeline

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Keep transformations reproducible

A **pipeline** records the sequence of transformations applied to the data.



Benefits

- **reproducible** experiments;
- fewer **manual errors**;
- consistent training and validation/test transformations;
- easier **deployment**;
- clearer separation of preprocessing and modeling steps.

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💡 A **pipeline** is not only software convenience. It helps preserve correct **experimental discipline**.

Worked Workflow Example

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Predicting apartment rental price

Features and preprocessing

Column	Type	Possible preprocessing
Area	Numerical	Scaling; check outliers
Number of rooms	Discrete numerical	Keep as count or scale
District	Nominal categorical	One-hot encoding
Building age	Numerical	Scaling; possible nonlinear terms
Furnished	Binary	Encode as 0/1
Monthly rent	**Target**	Exclude from input features

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Feature construction ideas

- Building age group
- Area per room
- Grouping rare districts
- Price per m^2 for descriptive analysis only

Avoid target leakage. Monthly rent and features derived from it, such as **price per m^2**, must not be used as inputs when predicting rent.

Worked Workflow Example

FCIS — Personal Use Only

Fraud detection features

Fraud detection requires features available **when the transaction is evaluated**.

Raw field	Possible feature	Leakage risk
Transaction amount	Scaled or log-transformed amount	**Low**
Merchant category	One-hot encoded category	**Low**
Country	Encoded country	**Low**
Previous failed attempts	Count before the transaction	**Low**, if available
Chargeback status	Exclude from inputs	**High**
Investigation result	Exclude from inputs	**High**
Previous transaction time	Time since last transaction	**Low**, if historical

Key caution. A feature may be informative but still **invalid for prediction** if it becomes available **only after the decision must be made**.

Part VIII

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Matching Preprocessing to the Model

Different learning algorithms have different requirements. **Preprocessing choices** should reflect how each model **learns from the data**.

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Linear and Logistic Regression

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Preprocessing priorities

Linear and logistic regression benefit from **appropriate feature preparation**, especially when regularization is used.

Preparing Features

- **Scale numerical features** when appropriate.
- **Encode categorical variables** meaningfully.
- Examine **correlated features** (multicollinearity).

Improving Model Flexibility

- Transform features to capture **nonlinear patterns**.
- Add **interaction features** when useful.
- Apply **regularization** to control complexity.

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Why does scaling matter?

With regularization, penalties depend on **coefficient magnitudes**. Features on very different scales can therefore be penalized **unequally**.

> 💡 **Key idea.** Coefficients for **income in SAR** and **age in years** are not directly comparable without considering their units and scales. Scaling is particularly important for **Ridge and Lasso**.

KNN and K-Means

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Distance-based methods are highly scale-sensitive

KNN and **K-Means** commonly rely on distances between observations, often using **Euclidean distance**:

$$d(x, z) = \sqrt{\sum_j (x_j - z_j)^2}$$

Example: Differences between two customers

Feature	Difference
Income	50,000
Age	5
Number of orders	3

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Without scaling, **income dominates the distance calculation** because its numerical difference is much larger.

> 💡 **Key idea.** Scaling is usually **essential for distance-based methods** when features have different units or ranges. Otherwise, distances may reflect **numerical scale rather than meaningful similarity**.

Decision Trees

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Less scale-sensitive, but not preprocessing-free

Decision trees learn rules by splitting observations according to feature values.

What Trees Handle Well

- Usually **do not require feature scaling**.
- Capture **nonlinear relationships**.
- Learn **decision rules** that can be interpreted.

What Still Needs Attention

- **Missing values**.
- **Categorical encoding**, depending on implementation.
- **Noisy or irrelevant features**.
- **Rare categories**.
- **Data leakage**.
- **Overfitting** from excessive tree depth.

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Important. A tree may not care whether age is measured in **years or standardized units**, but it can still **overfit noisy features** or learn misleading patterns from **leaked information**.

Neural Networks

FCIS — Personal Use Only

Clean numerical inputs support training

Neural networks typically benefit from:

- **Scaled numericals** feature.
- Consistent **categorical encoding**.
- **Correction** of invalid values.
- Appropriate **handling of missing data**.
- **Sufficient training data** for the model's complexity.
- **Validation monitoring** to detect overfitting.

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Why does scaling help?

Neural networks commonly use **gradient-based optimization**. Features with very different scales can make training **slower or less stable**.

💡 **Key idea.** Good preprocessing supports **stable and efficient training**. Combined with appropriate model complexity and validation, it can also improve **generalization**.

Part IX

FCIS — Personal Use Only

Practice, Diagnosis, and Common Mistakes

Effective **preprocessing** requires identifying data problems, choosing **appropriate solutions**, and avoiding common mistakes based on the **data, task, and model**.

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Activity: Diagnose Feature Types

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Decide the appropriate treatment

A dataset for predicting whether a customer will default contains:

Feature	Example value	Type	Possible treatment
Age	37		
Employment status	Employed		
Credit score	710		
Existing loan	Yes		
City	Madinah		
Application date	2026-03-14		
Default	No		

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Questions

1. Which column is the target?
2. Which features require encoding?
3. Which features may require scaling?
4. What useful features could be extracted from the date?

Decide the appropriate treatment

A dataset for predicting whether a customer will default contains:

Feature	Example value	Type	Possible treatment
Age	37		
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Default	No		

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Questions

1. Which column is the target?
2. Which features require encoding?
3. Which features may require scaling?
4. What useful features could be extracted from the date?

Possible reasoning: `Default` is the target; employment status, existing loan, and city need categorical encoding; age and credit score may be scaled; month, weekday, or time since previous application may be derived if meaningful.

Activity: Find the Leakage

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Which feature should not be used?

A hospital wants to predict whether a patient will be readmitted within 30 days.

Available columns:

Column	Description
Age	patient age at admission
Diagnosis	initial diagnosis
Length of stay	number of days in hospital
Discharge medication count	number of medications at discharge
Readmission date	date of next admission if any
Previous admissions	number of admissions before current one
Follow-up scheduled	whether follow-up was scheduled before discharge

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Questions

1. Which column is the most obvious leakage feature?
2. Which features depend on the prediction time?
3. What should be available when the model makes its prediction?

Which feature should not be used?

A hospital wants to predict whether a patient will be readmitted within 30 days.

Available columns:

Column	Description
Age	patient age at admission
Diagnosis	initial diagnosis
Length of stay	number of days in hospital
Discharge medication count	number of medications at discharge
Readmission date	date of next admission if any
Previous admissions	number of admissions before current one
Follow-up scheduled	whether follow-up was scheduled before discharge

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Questions

1. Which column is the most obvious leakage feature?
2. Which features depend on the prediction time?
3. What should be available when the model makes its prediction?

> `Readmission date` directly reveals the target and should not be used. Whether `length of stay` is valid depends on when the prediction is made: at admission or at discharge.

Activity: Choose a Transformation

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Match the problem to the preprocessing step

Situation	Candidate preprocessing
Numerical features have very different ranges	
Nominal variable has three categories	
Feature is strongly right-skewed	
Important variable is missing in 5% of records	
Target value appears as a hidden input feature	
A date field may contain useful seasonality	

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Match the problem to the preprocessing step

Situation	Candidate preprocessing
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Nominal variable has three categories	
Feature is strongly right-skewed	
Important variable is missing in 5% of records	
Target value appears as a hidden input feature	
A date field may contain useful seasonality	

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Possible answers: scaling; one-hot encoding; log transformation; imputation with justification; remove leaked feature; extract month/day/semester features.

Common Mistakes

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Avoid These Preprocessing Errors

Mistake	Why it matters
Fitting a scaler before splitting the data	**Validation/test information leaks** into training
Encoding categories inconsistently across splits	Feature columns may **no longer match**
Removing all outliers automatically	May discard **rare but important observations**
Treating nominal categories as ordered numbers	Introduces **artificial order and distances**
Ignoring why values are missing	May hide **informative patterns** or introduce bias
Including IDs without justification	May encourage **memorization instead of generalization**
Selecting features using test results	Compromises **independent test evaluation**
Not documenting preprocessing steps	Makes results **difficult to reproduce**

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Mini-Quiz

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Check your understanding

1. Why should we inspect a dataset before fitting a model?
2. What is the difference between a nominal and ordinal categorical variable?
3. Why is the median often preferred when a distribution has extreme values?
4. What does the IQR measure?
5. Why does KNN require careful feature scaling?
6. Give one example of data leakage.
7. Why should preprocessing parameters be estimated from the training set only?
8. What is one advantage and one disadvantage of one-hot encoding?

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Check your understanding

1. Why should we inspect a dataset before fitting a model?
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8. What is one advantage and one disadvantage of one-hot encoding?

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Suggested answers: to understand variables and data quality; nominal has no order while ordinal has meaningful order; median is robust to extremes; IQR is $Q_3 - Q_1$; KNN uses distance; using the target-derived field as input is leakage; validation/test must simulate unseen data; one-hot avoids false order but increases dimensionality.

Lecture 2 Summary

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Key Ideas to Retain

Data Understanding

- Identify **observations, features, targets, metadata**, and data types.
- Use **statistics and visualizations** to explore center, spread, shape, and relationships.
- The **mean and standard deviation** are useful, while the **median and IQR** are more resistant to extreme values.
- Use **histograms, boxplots, and scatter plots** to identify distributions, outliers, and associations.
- Use **cross-tabulations and Chi-square reasoning** to examine categorical associations.

Preprocessing and Features

- Handle **missing values, duplicates, invalid records**, and inconsistent categories carefully.
- Choose **scaling methods** based on feature distributions and model requirements.
- Apply **appropriate encoding** to categorical features.
- Use **feature construction and selection** to capture useful patterns and reduce unnecessary complexity.
- Prevent **data leakage** by fitting preprocessing on training data only and keeping test evaluation independent.

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End-of-Lecture Check

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Five questions to test your understanding

1. What information should a data dictionary contain, and why does timing matter?
2. How do mean, median, standard deviation, quartiles, and IQR describe a numerical feature differently?
3. Why should an outlier not be removed automatically?
4. When is one-hot encoding preferable to label encoding?
5. Why must scaling, imputation, and feature selection be fitted inside the training workflow?

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💡 **Preparation for next lecture:** Review linear equations, residuals, squared error, and the idea of minimizing a loss function. **Lecture 3** begins model construction with Linear Regression, Optimization, and Regularization.

Suggested Reading

FCIS — Private Use Only

Primary and supporting references

Primary reference

G. James, D. Witten, T. Hastie, and R. Tibshirani, *An Introduction to Statistical Learning with Applications in Python*, Springer, 2023.

- Chapter 2: Statistical Learning
- Review the role of inputs, outputs, model flexibility, training error, and test error.
- Use the data examples to connect exploratory analysis with later modeling.

Supporting references

A. Géron, *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow*, O'Reilly, 2022.

- Review the end-to-end ML project workflow, data cleaning, train/test splitting, and preprocessing pipelines.

K. P. Murphy, *Probabilistic Machine Learning: An Introduction*, MIT Press, 2022.

- Use selected introductory sections for a deeper view of data representations and probabilistic assumptions.

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> 💡 When reading, focus on how preprocessing choices change the representation available to the model.